

Mark schemes

Q1.

(a) $1 - e^{-\frac{1}{\mu}x} = \frac{1}{4}, \quad \frac{3}{4}$

Either.

M1

$$Q_1 = \mu \ln \left(\frac{4}{3} \right)$$

M1 for attempting either Q_1 or Q_3 .

M1A1

$$Q_3 = \mu \ln 4$$

A1

$$\text{IQR} = \mu \ln 3$$

CAO

A1

5

(b) $E(X^2) = \int_0^{\infty} \frac{1}{\mu} x^2 e^{-\frac{1}{\mu}x} dx$

Knowledge of formula.

M1

$$= \left[-x^2 e^{-\frac{1}{\mu}x} \right]_0^{\infty} + \int_0^{\infty} 2x e^{-\frac{1}{\mu}x} dx$$

Using integration by parts.

M1A1

$$= 0 + 2\mu \cdot \mu = 2\mu^2 \quad (\text{AG})$$

A1

4

(c) $\text{SD} = \sqrt{2\mu^2 - \mu^2} = \mu$

B1

$$\ln 3 > 1 \Rightarrow \text{SD} < \text{IQR}$$

M1A1

3

[12]

Q2.

(a) $\int_0^x \lambda e^{-\lambda x} dx = \left[-e^{-\lambda x} \right]_0^x$

M1

$$= 1 - e^{-\lambda x}$$

A1

2

(b) 0.0533, 0.0821 (Accept 0.0532)

B1B1

2

[4]

Q3.

(a) $F(x) = 1 - e^{-\lambda x}, x \geq 0$

$F(x) = 1 - e^{-\lambda x}$ B1B0

B1

$F(x) = 0, x < 0$

Dependent

B1

2

(b) $1 - e^{-\lambda x} = \frac{3}{4}$

For either Q_1 or Q_3

M1

$Q_3 = \frac{1}{\lambda} \ln 4$

m1 for attempting to solve for either Q_1 or Q_3

m1A1

$1 - e^{-\lambda x} = \frac{1}{4}$

$Q_1 = \frac{1}{\lambda} \ln \frac{4}{3}$

A1

$IQR = \frac{1}{\lambda} \ln 3$

AG

A1

5

(c) (i) $E(X^2) = \int_0^{\infty} \lambda x^2 e^{-\lambda x} dx$

Limits required

M1

$$= \left[-x^2 e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} 2x e^{-\lambda x} dx$$

A1

$$= \left[-\frac{2x}{\lambda} e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} \frac{2}{\lambda} e^{-\lambda x} dx$$

A1

$$= \left[-\frac{2}{\lambda^2} e^{-\lambda x} \right]_0^{\infty}$$

A1

4

$$= \frac{2}{\lambda^2}$$

AG

(ii) $\text{Var}(X) = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$

AG

B1

1

(d) (i) $\frac{1}{\lambda} \ln 3 = \frac{4}{\lambda^2}$

M1

$$\lambda = \frac{4}{\ln 3}$$

A1

2

(ii) $\text{IQR} \rightarrow 0 \text{ as } \lambda \rightarrow \infty$

E1

1

[15]

Q4.

(a) (i) $F(x) = 1 - e^{-\frac{x}{200}}$
May be quoted

$$P(X < 120) = 1 - e^{-0.6}$$

M1

$$= 0.451$$

AWRT

A1

2

(ii) $P(X > 160) = e^{-0.8}$

M1

$$= 0.449$$

AWRT

A1

2

(iii) $P(X < 160 | X > 120)$

$$= \frac{1 - [0.4512 + 0.4493]}{1 - 0.4512}$$

or

$$= P(X < 40)$$

M1

$$= 1 - e^{-0.2}$$

A1

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$$= 0.181$$

AWRT

A1

3

(b) $1 - e^{-\frac{m}{200}} = 0.5$

$$\Rightarrow e^{-\frac{m}{200}} = 0.5$$

M1

$$\Rightarrow m = \ln 0.5 \times (-200)$$

M1

$$= 139 \text{ hours}$$

AWRT

Q5.

(a) $E(X) = \int_0^{\infty} \lambda x e^{-\lambda x} dx$

Use of

M1

$$= \left[-x e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

Integrate by parts

A1

$$= \left[\frac{-e^{-\lambda x}}{\lambda} \right]_0^{\infty}$$

Correctly

A1

$$= \frac{1}{\lambda}$$

AG

A1

4

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(b) (i) $\frac{1}{\alpha} = \frac{1}{62.5} = 0.016$

M1A1

2

(ii) $F(t) = 1 - e^{-0.016 \times 80}$

Or $\int_{80}^{\infty} 0.016 e^{-0.016t} dt$

Award M1A1 for complement

M1A1

$$P(T > 80) = e^{-0.016 \times 80}$$

M1

$$= 0.278$$

A1

4

(iii) **Either**

$$e^{-0.016 \times 100} + e^{-0.016 \times 80}$$

M1A1

$$= 0.726$$

A1

or

$$e^{-0.016 \times 20}$$

(M2)

$$= 0.726$$

(A1)

3

[13]

Q6.

(a) (i)

$$\left[-e^{-0.2x} \right]_0^2$$

M1

$$= 1 - e^{-0.4} = 0.330$$

AWRT (Accept 0.33)

A1

2

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(ii)

$$1 - \left[-e^{-0.2x} \right]_0^3$$

M1

$$= e^{-0.6} = 0.549$$

AWRT

A1

2

(b) (i)

$$E(X) = \int_0^{\infty} \lambda x e^{-\lambda x} dx$$

Limits Required

M1

$$= \left[-x e^{-\lambda x} \right]_0^{\infty} + \int_0^{\infty} e^{-\lambda x} dx$$

A1

$$= [0] - [0] + \left[\frac{-e^{-\lambda x}}{\lambda} \right]_0^{\infty}$$

A1

$$= 0 - \left[-\frac{1}{\lambda} \right]$$

$$= \frac{1}{\lambda}$$

A1

4

(ii) $\int_0^m \lambda e^{-\lambda x} dx = \frac{1}{2}$

Must use integration, not quote cdf

M1

$$\Rightarrow \left[-e^{-\lambda x} \right]_0^m = \frac{1}{2}$$

$$\Rightarrow 1 - e^{-\lambda m} = \frac{1}{2}$$

$$\Rightarrow e^{-\lambda m} = \frac{1}{2}$$

Dependent on previous M

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m1

$$\Rightarrow m = \frac{1}{\lambda} \ln 2$$

OE, eg $\frac{-\ln 0.5}{\lambda}$ or $\frac{0.693}{\lambda}$

A1

3

(c) (i) $\frac{1}{0.3} \times 24 = 80 \text{ hours}$

B1

1

$$\frac{1}{0.3} \ln 2 \times 24 = 80 \ln 2 \text{ hours}$$

Or 55.5 hours AWRT
M1 for 2.31 (days)

M1 A1

2

[14]

Q7.

$$(a) \quad (i) \quad f(t) = \begin{cases} 0.2e^{-0.2t} & t \geq 0, \\ 0 & \text{otherwise} \end{cases}$$

$$0.2 e^{-0.2t}$$

B1

$t \geq 0$ and zero range

$$(ii) \quad f(t) = \int_0^t 0.2e^{-0.2x} dx$$

integrates pdf

B1

$$= \left[-e^{-0.2x} \right]_0^t = 1 - e^{-0.2t}$$

integration and limits

M1

$$(iii) \quad P(2 \leq T \leq 8) = F(8) - F(2)$$

A1

2

$$= (1 - e^{-1.6}) - (1 - e^{-0.4})$$

M1

$$\text{or } e^{-0.4} - e^{-1.6}$$

A1

$$= 0.7981 - 0.3297$$

$$= 0.468$$

AWRT; CAO

A1

3

$$(b) \quad (i) \quad P(T > 10) = 1 - F(10)$$

M1

$$= e^{-2} = 0.135$$

AWRT; CAO

A1

2

$$(ii) \quad 0.15 + 0.85 \times P(T \leq 5) \\ \text{or } 1 - 0.85 \times P(T > 5)$$

M1

$$= 0.15 + 0.85(1 - e^{-1}) \\ \text{or} \quad = 1 - 0.85 \times e^{-1}$$

A1

$$= 0.687 \\ \text{or} \quad = 0.687 \\ \text{AWRT}$$

A1

3

[12]

Q8.

$$(a) \quad F(t) = \int_0^t \frac{1}{2} e^{-\frac{x}{2}} dx$$

Integration and limits

M1

$$= \left[-e^{-\frac{x}{2}} \right]_0^t$$

Correct integration

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A1

$$= 1 - e^{-\frac{t}{2}}$$

Printed answer

2

$$(b) \quad (i) \quad P(T \leq 3) = F(3) \\ = 1 - e^{-1.5} \\ \text{or by integration}$$

M1

$$= 0.777 \\ \text{AWFW } 0.776 \text{ to } 0.777$$

A1

2

$$(ii) \quad \text{We require } P(T > 1)$$

Could be implied

M1

$$= 1 - F(1)$$

m1

$$= e^{-0.5} = 0.607$$

AWFW 0.606 to 0.607

A1

3

(iii) We require $P(T \leq 3 | T > 1)$

Could be implied

M1

$$= \frac{P(1 \leq T \leq 3)}{P(T > 1)}$$

$$= \frac{F(3) - F(1)}{1 - F(1)}$$

or by integration

M1

$$= \frac{(1 - e^{-1.5}) - (1 - e^{-0.5})}{e^{-1.5}}$$

*or uses answers to (i) and (ii)
ft on previous answers*

A1f.t.

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$$= 0.632$$

AWFW 0.630 to 0.634

B1 for just $F(3) - F(1)$

A1

4

(c) $X = 3 + T$

Could be implied

M1

$$\text{Mean} = 3 + 2 = 5$$

CAO

A1

$$\text{Standard deviation} = 2$$

CAO

A1

3

[14]

Q9.

- (a) Shape of histogram similar to pdf of exponential distribution.

E1

Mean and SD approximately equal.

E1

2

(b) (i) $E(T) = \frac{1}{0.3} = 3.33$
awrt

B1

1

(ii) $P(T \leq 1) = F(1)$

M1

$= 1 - e^{-0.3}$
 $= 0.259$
 awrt

A1

2

(iii) $P(T > 1.75 \mid T > 1)$
identifies correct probability.

M1

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$= \frac{1 - F(1.75)}{1 - F(1)} \left(= \frac{1 - F(1.75)}{0.741} \right)$
 numerator correct

a1

$= \frac{e^{-0.525}}{e^{-0.3}} = 0.799$

f.t. on answer to (b)(ii)
 B1 for $P(T < 1.75 \mid T > 1)$ correctly evaluated.

A1f.t.

3

(iv) Let median value be m
 $F(m) = 0.5$
 $1 - e^{-0.3m} = 0.5$

M1

$$e^{-0.3m} = 0.5$$

$$-0.3m = \ln(0.5)$$

valid attempt to solve

m1

$$m = 2.31$$

cao

A1

Median time interval = 2.31 minutes

3

[11]

Q10.

(a) (i) $f(t) = \frac{1}{10} e^{-\frac{t}{10}} \quad t > 0$

CAO

$f(t) = 0 \quad t < 0$

CAO 0 & both ranges

B1

B1

2

(ii) $F(t) = \int_0^t \frac{1}{10} e^{-\frac{x}{10}} dx =$

use of $\int f(x) dx$

condone use of t

M1

$$\left[-e^{-\frac{x}{10}} \right]_0^t = 1 - e^{-\frac{t}{10}}$$

integral & limits

CAO

A1A1

3

(b) (i) Required $P(T < 15) = F(15) =$

use of $F(15)$, or integration

M1

$$0.776 \text{ to } 0.777$$

AWFW

A1

2

- (ii) Require $P(10 < T < 20) = F(20) - F(10) =$
use of difference of Fs or integration

M1

$$0.86466 - 0.63212 = 0.232 \text{ to } 0.233$$

AWFW

A1

2

[9]

Q11.

(a) $f(x) = \frac{1}{10} e^{-\frac{x}{10}}$

B1

For $x > 0$,

$$F(x) = \frac{1}{10} \int_0^x e^{-\frac{t}{10}} dt = \left[-e^{-\frac{t}{10}} \right]_0^x$$

M1A1 A1

4

= AG

(b) $P(X > 15) = e^{-\frac{15}{10}} = 0.223$

M1A1

2

(c) $P(X < 20) | X > 15 = \frac{P(15 < X < 20)}{P(X > 15)}$

M1

$$= \frac{e^{-\frac{15}{10}} - e^{-\frac{20}{10}}}{e^{-\frac{15}{10}}}$$

A1

$$= 0.393$$

Accept 0.394

A1

3

[9]

Q12.

(a) (i) $f(t) = \frac{1}{800} e^{-\frac{t}{800}}$
CAO

B1

$f(t) = 0$ for $t \geq 0$
for $t < 0$
CAO both

B1

2

(ii) $\sigma = 800$
CAO

B1

1

(iii) $P(T > +1000) = \int_{1000}^{\infty} \frac{1}{800} e^{-\frac{t}{800}} dt$
Use of $\int f(t) dt$

M1

$= \left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}$
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integral & limits

A1

$= e^{-\frac{1000}{800}} = 0.286 \text{ to } 0.287$
AWFW

A1

3

(iv) $P(T > +1000 | T > 1000) =$
Use of $P(A \& B) / P(A)$

M1

$\frac{P(T > 2000)}{P(T > 1000)} =$

OE

A1

$$\frac{\left[-e^{-\frac{t}{800}} \right]_{2000}^{\infty}}{\left[-e^{-\frac{t}{800}} \right]_{1000}^{\infty}} = 0.286 \text{ to } 0.287$$

AWRT

If answer (iv) = answer (iii), then 3 marks
if answer (iii) correct, otherwise 2 marks

A1

3

- (b) Number of components working after 1000 hours ~ binomial

M1

with $n = 4$ and $p = 0.286/7$
f.t. on (iii)

A1f.t.

$P(X \geq 2) = P(X = 2, 3, 4) =$
use of either

M1

$1 - P(X = 0, 1) =$

0.323 to 0.326

AWFW

A1

4

[13]

Q13.

$T \sim E(2)$

- (a) 1

cao; accept 'unity'

B1

1

- (b) $P(S > 5) = P(T > 4)$
4 cao

B1

$$= 1 - \left(1 - e^{-\frac{4}{2}} \right) = e^{-2}$$

*use of exponential cdf or pdf
with $\lambda = 0.5$ or 2*

M1

$$= 0.135$$

awrt

A1

3

(c) $P(S < 5 \mid S > 3) = P(T < 4 \mid T > 2)$

use of conditional probability

M1

Exponential has 'no memory' so
use of; may be implied

$$= P(T < 2)$$

2 cao; (even from 5 - 3)

M1

$$= 1 - e^{-1} = 0.632$$

awrt

A1

A1

4

(d) Probability = $(b)^5 = 0.000044$ to 0.000046 . All rights reserved.
awfw; ft on (b)

B1ft

Implies an extremely rare event
rare event, or equivalent

E1

2

so casts doubt on model

ag

[10]

Q14.

$$D \sim \text{Exp}(125)$$

$$(a) \quad P(D < 100) = \left[-e^{-\frac{d}{125}} \right]_0^{100} = 1 - e^{-\frac{100}{125}}$$

use of PDF of DF

M1

$$= 1 - e^{-0.8} = 0.550 \text{ to } 0.551$$

awfw; accept 0.55

A1

2

$$(b) \quad P(100 < D < 300) = P(D < 300) - (a)$$

use of; oe

M1

$$(1 - e^{-2.4}) - (1 - e^{-0.8}) = e^{-0.8} - e^{-2.4}$$

correct expression; oe

A1ft

$$= 0.909 - 0.551 = 0.449 - 0.091$$

ft on (a)

$$= 0.358 \text{ to } 0.359$$

awfw

A1

3

[5]

EXAM PAPERS PRACTICE

Q15.

$$(a) \quad SD(T) = 8$$

cao

B1

1

$$(b) \quad P(5 < T < 15) = P(T < 15) - P(T < 5)$$

$$\text{Use of; or use of } \int \lambda e^{-\lambda t} dt$$

M1

$$= \left(1 - e^{-\frac{15}{8}} \right) - \left(1 - e^{-\frac{5}{8}} \right)$$

$$\text{Or } \left[-e^{-\lambda t} \right]_5^{15}$$

≥ 1 correct term

A1

$$= e^{-\frac{5}{8}} - e^{-\frac{15}{8}}$$

$$= 0.53526 - 0.15335 = 0.382$$

awrt

A1

3

(c) $P(\text{none from 5.45 to 6.00}) = P(T > 15)$

Attempt at either probability

M1

$$= e^{-\frac{15}{8}} = 0.153$$

f.t. on part (b) (0.153355)

A1f.t.

$P(\text{none from 6.00 to 6.15}) = P(S > 15)$

$$= e^{-1} = 0.368$$

cao/awrt

(0.367879)

A1

$P(\text{none from 5.45 to 6.15}) = \text{'product'}$

Use of

m1

$$= e^{-\frac{23}{8}} = 0.056 \text{ to } 0.057$$

awfw

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A1

5

Or (using Poisson) =

(M1)

$$P(0 \mid \lambda = \frac{15}{8} = 1.875)$$

(A1)

×

$$P(0 \mid \lambda = \frac{15}{15} = 1)$$

(A1)

$$= e^{-1.875} \times e^{-1}$$

Product

(m1)

$$= e^{-2.875} \times 0.056 \text{ to } 0.057$$

awfw

(A1)

[9]

Q16.

(a) $D \sim \text{Exp}(16)$

Standard deviation, $\sigma = 16$

cao

B1

1

(b) (i) Exponential with $\frac{1}{\lambda} = 16$
Use of associated pdf or df

M1

$$P(D > 20) = \left[-e^{-\frac{d}{16}} \right]_{20}^{\infty}$$

or

$$= 1 - \left(1 - e^{-\frac{20}{16}} \right)$$

Either

A1

$$e^{-1.25} = 0.286 \text{ to } 0.287$$

awfw

A1

(ii) $P(20 < D < 30) = e^{-1.25} - e^{-1.875}$

awfw

Difference, or equivalent

A1

$$= 0.28650 - 0.15335 = 0.133$$

awrt

A1

5

(c) $P(D_2 > 20 | D_1 > 20) =$

Use of conditional probability

M1

$$\frac{P(D_1 + D_2 > 20 | D_1 > 20)}{P(D_1 > 20)}$$

Correct expression

A1

$$e^{-25} \div e^{-125} = e^{-125} = 0.286 \text{ to } 0.287$$

ft if calculated answer = (b)(i)

Justification of equality $\Rightarrow$ 3 when (b)(i)

correct

Justification of equality $\Rightarrow$ 2 when (b)(i)

incorrect

A1.ft

3

(d) Poisson

B1

$$\text{Parameter/mean} = \frac{5}{16}$$

$$= 0.312 \text{ to } 0.13$$

cao/awfw

B1

2

EXAM PAPERS PRACTICE [11]

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Q17.

(a) $P(X \leq x) = 1 - e^{-x/6}$

$$P(X < 3) = 1 - e^{-3/6}$$

use of 3 F(x)

M1

$$= 1 - e^{-0.5}$$

$$= 1 - 0.6065 = \underline{0.39}$$

awrt

A1

2

(b) $P(5 < X < 10)$

$$= P(X < 10) - P(X < 5)$$

use of difference

M1

$$= (1 - e^{-10/6}) - (1 - e^{-5/6})$$

use of 5 and 10 in $F(x)$

M1

$$= e^{-5/6} - e^{-10/6}$$

$$= 0.4346 - 0.1889$$

$$= 0.24 \text{ to } 0.25$$

awfw

A1

3

[5]



EXAM PAPERS PRACTICE

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Examiner reports

Q1.

Students were far happier with this question, where their pure mathematics ability enabled them to deal well with logarithms and the integration that was required to answer this question. Almost all seemed familiar with the statistical knowledge required. It was noticed however that 'range' was sometimes used when 'interquartile range' was meant. Since both are well-defined concepts, with distinctly different meanings, this could not be accepted. Otherwise, the few marks that were lost resulted from small matters of detail, for example sign errors, or misuse of brackets. It was pleasing to note that the vast majority of students realised that $3 > 1$ was needed to answer the last part of the question correctly.

Q2.

This question was answered better than any other question on the paper, with many candidates scoring full marks. The procedures were well understood and arithmetical accuracy was meticulous. Only the occasional candidate forgot to combine classes, where the expected values fell below 5, or looked up the wrong critical value of chi-squared from the table.

Q3.

The mathematical skills of most candidates were well able to cope with the pure mathematics involved in this question. In part (a), a lack of detail was evident in the answers. To gain both marks it was necessary to state $F(x) = 1 - e^{-\lambda x}$ for $x \geq 0$ and $F(x) = 0$ otherwise; simply stating $F(x) = 1 - e^{-\lambda x}$ only obtained 1 mark.

Many candidates were able to find the two quartiles and hence the interquartile range in part (b). In part (c)(i), limits were required for the integration in order to score marks and, in fact, most candidates did use definite integration. As was the case in question 4, some lost the final mark for not showing a complete solution. Part (d)(i) was usually done successfully by using the printed answers but, in part (d)(ii), some candidates lost the final mark for an imprecise statement such as "it gets smaller".

Q4.

Parts (a)(i) and (a)(ii) were answered correctly by the majority of candidates. The few errors that did occur usually involved finding the complement of the required probability. Part (a)(iii) caused major difficulties to many candidates. Nearly all such candidates did not divide by $P(X > 120)$ and/or did not include all relevant probabilities in the numerator. In part (b), many candidates found difficulty in establishing the correct equation. However, some were able to earn a method mark for solving their equation using exponentials and logarithms correctly.

Q5.

Part (a) produced considerable variation in the responses. Weaker candidates used incorrect limits, or omitted the limits completely, and sometimes candidates showed little knowledge of integration by parts. The more able candidates showed impressive knowledge of how to handle infinite integrals. As this is an applied unit, a somewhat informal approach, without the rigours of pure mathematics, to infinite integrals is acceptable. The specification requires candidates to know the form of the cumulative distribution function for the exponential distribution. Those who did were able to carry out

the calculations in parts (b)(ii) and (iii) without any great difficulty. Again, the ability to deal with conditional probability was evident in part (b)(iii).

Q6.

This was one of the questions that only most able candidates were able to do completely correctly. Most candidates obtained accurate answers in part (a). However, in part (b), the limits for integrals were frequently either omitted or incorrect. In part (b)(ii), it was necessary to 'use integration' as the question required and not merely to quote a formula. Too many candidates did not multiply by 24 to get answers in hours in part (c).

Q7.

This question caused some difficulty. The request to specify fully the probability density function in part (a)(i) required inclusion of $f(t) = 0$ for $t < 0$. The remainder of part (a) was well answered. In part (b), the need to consider whether or not customers were waiting when the snack bar opened and to apply laws of probability proved challenging.

Q8.

In general, this question produced some excellent work. Candidates coped well with the combination of a fixed time interval and an exponentially distributed time interval. They made efficient use of the formula obtained in part (a) to find the probabilities asked for in part (b).

Q9.

Many candidates answered part (a) clearly and concisely and scored full marks. Others ignored the request to use the given summary information and simply described the properties of an exponential distribution; this did not gain any credit.

Part (b) was generally well answered except for part (ii) where few candidates were able to identify the required conditional probability correctly. A small minority of candidates treated the given cumulative distribution function as a probability density function and made little progress.

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Q10.

This question was well answered. Candidates remembered to state the range ($t \geq 0$) for

which the formula $f(t) = \frac{1}{10} e^{-\frac{t}{10}}$ was valid and also to state that $f(t) = 0$ for $t < 0$. In part (b) candidates favoured the method of integration, rather than using the distribution function $F(t)$, as a probability formula for $P(T \leq t)$.

Q12.

For some candidates this was not a good first question but this certainly did not appear to affect their subsequent attempts at other questions. In part (a)(i), many candidates omitted the range $t < 0$ and/ or 0 for $t < 0$ and, in part (a)(ii), some candidates quoted 800^{-1} for the standard deviation. Nevertheless, this did not prevent them, and indeed almost all candidates, from scoring the 3 marks available in part (a)(iii). In part (b)(iv), only the most able candidates appeared to be aware that, for an exponential distribution, $P(T > t_2 | T > t_1) = P(T > t_2)$ but some other candidates were able to make significant progress, even if not with complete success, by the application of $P(A | B) = P(A \cap B) \div P(B)$. Whilst many

candidates realised that part (b) required their answer to part (a)(ii), only the most able candidates also realised that a binomial distribution was then needed to evaluate $P(X = 2, 3, 4)$ or $1 - P(X = 0, 1)$

Q13.

Answers to this question were generally very poor. At least 50% of candidates used 0.5 for λ rather than 2 and so worked with e^{-2t} . Although these, and most other, candidates were able to answer part (a) correctly, they made somewhat of a mess of part (b) often compounding the error by not changing 5 to 4. Given these errors, correct answers to part (c) were extremely rare with the vast majority of candidates simply attempting $P(3 < T < 5)$.

In part (d), most candidates made some general statement about trains in stations and failed to realise that reference was expected to part (b) and the value of $(P(T > 4))^5$.

Q14.

The majority of candidates scored full marks on this question by correct application of the correct distribution function; integration of the probability density function was very rare. A minority of candidates found complementary answers to one part, whilst a much smaller

number tried working with $1 - e^{-\frac{125}{4}}$. Thankfully no attempts at continuity corrections were seen.

Q15.

In part (a), better candidates stated “8” but the majority stated “ $\sqrt{8}$ ” or “0.125.” However part (b) was answered correctly by the vast majority of candidates. Sadly, the same was not the case in part (c). Candidates often treated the interval as one 30-minute period with $\lambda = 30$ (2×15) or 23 ($8 + 15$). Some candidates even tried the direct substitution of 5.45 and 6.15 into their exponential distribution function or attempted a normal linear combination. Even when two correct probabilities were calculated, they were all too often added rather than multiplied.

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Q16.

It was disappointing to see the large proportion of candidates who could not score the first mark despite the relevant information being stated clearly on page 14 of the formulae booklet. Nevertheless, most candidates scored most, often all, of the 5 marks available in part (b). In part (c), very few candidates appeared aware of the ‘non-memory’ property of an exponential distribution. Of the few who scored marks, less than half quoted their answers to part (a)(i) with most starting a new using conditional probability with varying degrees of success. In part (d), about 50% of candidates scored 1 mark for either

“Poisson” or “ $\frac{5}{16}$ ”, but very few indeed scored the 2 marks available.

Q17.

Apart from the significant minority who ignored the word ‘thousand’ on the first line of the question, this was often a source of another 5 marks. A few candidates confused areas in part (b), often when trying to subtract the two tail areas from unity.